

The differential cross-section for coherent vector meson production:

$$\frac{d\sigma^{\gamma^*+A \rightarrow V+A}}{d|t|} = \frac{1}{16\pi} |\langle \mathcal{A} \rangle_\Omega|^2 \quad (1)$$

Note

$$d|t| = d|\mathbf{\Delta}_\perp|^2 = 2|\mathbf{\Delta}_\perp| d|\mathbf{\Delta}_\perp| = \frac{1}{\pi} d^2 \mathbf{\Delta}_\perp \quad (2)$$

Then we can write the differential cross-section as

$$\frac{d\sigma^{\gamma^*+A \rightarrow V+A}}{d^2 \mathbf{\Delta}_\perp} = \frac{1}{16\pi^2} |\langle \mathcal{A} \rangle_\Omega|^2 \quad (3)$$

$$\sigma^{\gamma^*+A \rightarrow V+A} = \frac{1}{16\pi^2} \int d^2 \mathbf{\Delta}_\perp |\langle \mathcal{A} \rangle_\Omega|^2 \quad (4)$$

Recall

$$\mathcal{A} = 2i \int d^2 \mathbf{r}_\perp d^2 \mathbf{b}_\perp \frac{dz}{4\pi} e^{-i\mathbf{b}_\perp \cdot \mathbf{\Delta}_\perp} [\Psi_V^* \Psi_\gamma](Q^2, \mathbf{r}_\perp, z) N_\Omega(\mathbf{r}_\perp, \mathbf{b}_\perp, z, x) \quad (5)$$

where¹

$$N_\Omega(\mathbf{r}_\perp, \mathbf{b}_\perp, z, x) = 1 - \frac{1}{N_c} \text{Tr} [V(\mathbf{b}_\perp + (1-z)\mathbf{r}_\perp) V^\dagger(\mathbf{b}_\perp - z\mathbf{r}_\perp)] \quad (6)$$

Let us define

$$F(\mathbf{b}_\perp, x) = \int d^2 \mathbf{r}_\perp dz [\Psi_V^* \Psi_\gamma](Q^2, \mathbf{r}_\perp, z) N_\Omega(\mathbf{r}_\perp, \mathbf{b}_\perp, z, x) \quad (7)$$

Then

$$\mathcal{A} = i \int \frac{d^2 \mathbf{b}_\perp}{2\pi} e^{-i\mathbf{b}_\perp \cdot \mathbf{\Delta}_\perp} F(\mathbf{b}_\perp, x) \quad (8)$$

Then

$$|\langle \mathcal{A} \rangle_\Omega|^2 = \int \frac{d^2 \mathbf{b}_\perp}{2\pi} e^{-i\mathbf{b}_\perp \cdot \mathbf{\Delta}_\perp} \langle F(\mathbf{b}_\perp, x) \rangle_\Omega \int \frac{d^2 \mathbf{b}'_\perp}{2\pi} e^{i\mathbf{b}'_\perp \cdot \mathbf{\Delta}_\perp} \langle F^*(\mathbf{b}'_\perp, x) \rangle_\Omega \quad (9)$$

The integral over $\mathbf{\Delta}_\perp$ is trivial as it only appears on the phase giving a $(2\pi)^2 \delta^{(2)}(\mathbf{b}_\perp - \mathbf{b}'_\perp)$, we then integrate over $\mathbf{b}'_\perp$

$$\int d^2 \mathbf{\Delta}_\perp |\langle \mathcal{A} \rangle_\Omega|^2 = \int d^2 \mathbf{\Delta}_\perp \int \frac{d^2 \mathbf{b}_\perp}{2\pi} e^{-i\mathbf{b}_\perp \cdot \mathbf{\Delta}_\perp} \langle F(\mathbf{b}_\perp, x) \rangle_\Omega \int \frac{d^2 \mathbf{b}'_\perp}{2\pi} e^{i\mathbf{b}'_\perp \cdot \mathbf{\Delta}_\perp} \langle F^*(\mathbf{b}'_\perp, x) \rangle_\Omega \quad (10)$$

$$\begin{aligned} &= \int d^2 \mathbf{b}_\perp \langle F(\mathbf{b}_\perp, x) \rangle_\Omega \langle F^*(\mathbf{b}_\perp, x) \rangle_\Omega \\ &= \int d^2 \mathbf{b}_\perp |\langle F(\mathbf{b}_\perp, x) \rangle_\Omega|^2 \end{aligned} \quad (11)$$

Finally, we have

$$\sigma^{\gamma^*+A \rightarrow V+A} = \frac{1}{16\pi^2} \int d^2 \mathbf{b}_\perp |\langle F(\mathbf{b}_\perp, x) \rangle_\Omega|^2 \quad (12)$$

¹ Off-forward phase has been absorbed into the dipole coordinates

where

$$F(\mathbf{b}_\perp, x) = \int d^2\mathbf{r}_\perp dz [\Psi_V^* \Psi_\gamma](Q^2, \mathbf{r}_\perp, z) N_\Omega(\mathbf{r}_\perp, \mathbf{b}_\perp, z, x) \quad (13)$$

Event-by-event $F(\mathbf{b}_\perp, x)$ might depend on the angle of $\mathbf{b}_\perp$ but after average it only depends on $|\mathbf{b}_\perp|$, so for a given x , $\langle F(|\mathbf{b}_\perp|, x) \rangle_\Omega$ can be stored in a one-dimensional table. $\langle F(|\mathbf{b}_\perp|, x) \rangle_\Omega$ should be a smooth function of $|\mathbf{b}_\perp|$, it can be understood as an x -dependent "effective" thickness function ².

$$\sigma^{\gamma^*+A \rightarrow V+A} = \frac{1}{8\pi} \int |\mathbf{b}_\perp| d|\mathbf{b}_\perp| |\langle F(|\mathbf{b}_\perp|, x) \rangle_\Omega|^2 \quad (14)$$

Then the total coherent cross-section is computed from a one-dimensional integral over $|\mathbf{b}_\perp|$.

In order to compute the incoherent cross section, one has to store $F(|\mathbf{b}_\perp|, \theta)$ for every configuration and compute The incoherent cross-section is

$$\sigma^{\gamma^*+A \rightarrow V+A^*} = \frac{1}{16\pi^2} \int d^2\mathbf{b}_\perp \left[\langle |F(\mathbf{b}_\perp, x)|^2 \rangle_\Omega - |\langle F(\mathbf{b}_\perp, x) \rangle_\Omega|^2 \right]. \quad (15)$$

² In the dilute limit in which $\langle N_\Omega(\mathbf{r}_\perp, \mathbf{b}_\perp, z, x) \rangle_\Omega = \langle N_\Omega(\mathbf{r}_\perp, x) \rangle_\Omega T(\mathbf{b}_\perp)$, then $\langle F(\mathbf{b}_\perp, x) \rangle_\Omega \propto T(\mathbf{b}_\perp)$